HiWatch OPC UA data communication
User documentation
V1.4

Table of Contents

1 SCOPE **3**

2 GENERAL INFORMATION **3**

3 USAGE **3**

3.1 REMOTE COMMAND AND CONTROLS 3
3.2 CONNECTION SECURITY 4

4 OPC UA NODE STRUCTURE **4**

4.1 ROOT LEVEL NODES 4
4.2 DATA NODE FOLDERS 4
4.2.1 Data node status codes *5*
4.3 OPC UA SERVER NODE STRUCTURE REFERENCE 5
4.3.1 ROOT FOLDER *5*
4.3.2 FOLDER ID: ns=2;s=Sensor_00 ..... *6*
4.3.3 FOLDER ID: ns=2;s=Sensor_00.CurrentMean *6*
4.3.4 FOLDER ID: ns=2;s=Sensor_00.CurrentParticles *7*
4.3.5 FOLDER ID: ns=2;s=Sensor_00.PlumeStatistics *7*
4.3.6 FOLDER ID: ns=2;s=Sensor_00.SensorStatus *8*
4.3.7 FOLDER ID: ns=2;s=Sensor_00.SessionData *9*
4.3.8 FOLDER ID: ns=2;s=Sensor_00.OPERATIONS *10*
4.3.9 FOLDER ID: ns=2;s=Sensor_00.CurrentMean *11*
4.4 NOTES FOR SPECIFIC DATA NODES: 12
4.5 ALERT OBJECTS 12

5 COMMAND INTERFACE **12**

5.1 COMMAND STRUCTURE 12
5.2 AVAILABLE COMMANDS 12
5.2.1 SENSOR *13*
5.2.2 PULSE *13*
5.2.3 SESSION *13*

LIMITATIONS **14**

6 INSTALLING HIWATCH OPC UA HMI SOFTWARE **15**

6.1 RUNNING THE HIWATCH OPC UA INSTALLER 15
6.2 ACTIVATION OF OPC UA SERVER LICENSE 19
6.3 RUNNING HIWATCH OPC UA SERVER 20

1 Scope

This document is related to OPC UA data communication protocol as defined with HiWatch class spray particle sensors.

OPC UA node tree structure is presented with usage description.Contents of individul nodes are described. Use of HiWatch sensor command interface is defined.

2 General information

HiWatch sensor control software includes OPC UA server starting from Version 3.5 as an optional component. Running the server requires a National Instruments software license:

LabView OPC UA Toolkit deployment license, part no: 785292-35

One license per server computer is required. The installation of the OPC UA server components must be enabled in the main software installer. Normal shipment of the software includes a single permanent license. When shipped with a complete sensor system, the license is pre-installed on the sensor controller and is non-transferable.

The presence of the OPC UA server functionality can be verified in the main window title sting, and by the presence ot the Remote tab in camera control interface. The software version identifier contains the string ”OPC_UA” if the functionality is included.

3 Usage

The OPC UA server process is started automatically when the HiWatch control software is run. Sensor nodes are populated only after the camera head is connected and initialized with a measurement profile.

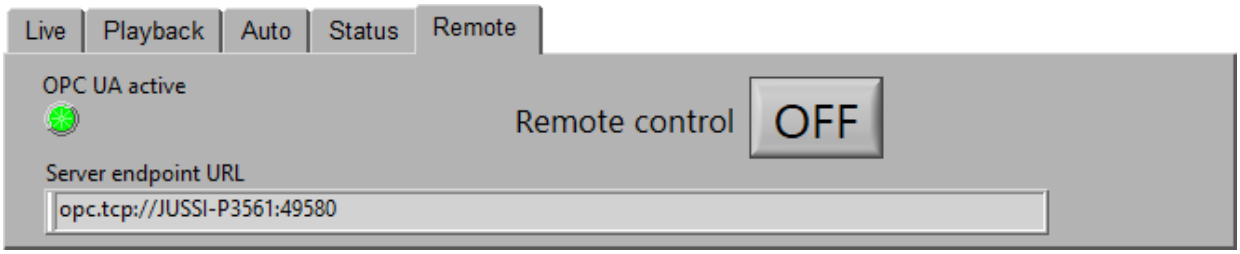


Figure 1: OPC UA server connection details shown in HiWatch software. Host name in the URL should be substituted with the actual hostname,

OPC UA server accepts local and remote connections using OPC.TCP protocol. Other protocols are not supported. The URL for the server connection is displayed on the **Remote** tab in the device control section (Fig. 1). The user must arrange the network connection between OPC server and remote client, TCP/IP routing and name resolution in order to use remote clients.

If a network firewall is present on the server computer, TCP port 49580 must be open for incoming connections from the network to enable remote connections.

NOTE: TCP connections with the server may be interrupted when the sensor is connected or disconnected, or when the alert limit settings are changed. For availability reasons, enforcing automatic reconnection of the clients is encouraged.

3.1 Remote command and controls

HiWatch OPC UA publishes the data of sensor status and measurement results over the OPC.TCP connection. The user may also enable remote command and control (C&C) operation. C&C is enabled by checking **Remote control** button in the tab.

The remote C&C mode has the following features:

- control of camera and laser status is exclusively by the remote connection
- camera is switched to Standby/Idle state when the mode is entered
- Laser trigger is enabled
- Analysis mode is activated

In remote operation, the status of the sensor should be monitored continuously over the TCP connection. CameraState, CameraTemp, FrameRate and SensorErrorCode are indicators of possible issues in sensor operation. Any error condition of the sensor must be handled in the local interface.

Refer to Section 5 for more information about the remote control implementation.

3.2 Connection security

HiWatch OPC UA server uses only plaintext communication. No encrypted connection is available in the present version. No user identification is made for the connection. Using an isolated network between server and client is strongly recommended if any network security issues are relevant.

4 OPC UA Node structure

The node tree structure offered by the HiWatch OPC UA server is presented in figure 2. All nodes are client readable, only COMMAND node is client writable. The details for each node are explained in the Node Reference section.

4.1 Root level nodes

The HiWatch OPC UA server has root level data nodes that identify the software that runs the server.

- SoftwareClass
- SoftwareVersion

Refer to these nodes to resolve any compatibility issues with client software. This manual covers SoftwareVersion 3.5.

4.2 Data node folders

The sensor data output is provided by the following node folders:

- CurrentMean: latest filtered mean data values from the sensor. These correspond to the items in the HiWatch HMI History page.
- CurrentParticles: data arrays that contain individual particle data from the most recent image bundle analyzed by the software. This data can be assembled and compiled to form the equivalent of various data distributions in the HiWatch HMI.
- PlumeStatistics: scalar data vaues complied from the current Particle dataset contained in the HMI. They represent data collected over a longer period of time, as specified in the Particle Collection bar of the HMI.
- SensorStatus: various items related to the physical state of the sensor. Data collection clients should monitor these nodes to be aware of the status of the sensor operation.
- SessionData: various items related to the remote controlled data collection. Any client using the remote control functions should monitor the related

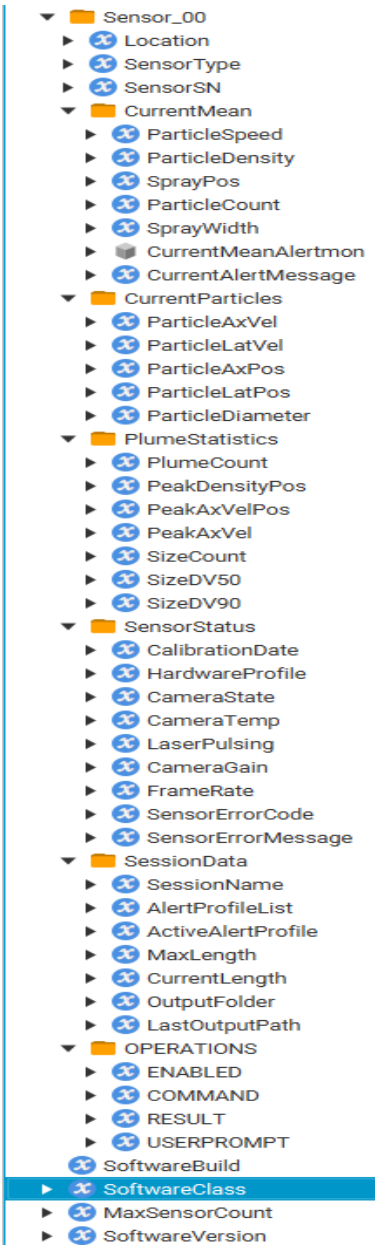


Figure 2: HiWatch CS OPC UA node tree

nodes to verify the status of the data collection.

4.2.1 Data node status codes

OPC UA status codes are used to monitor the quality of collected data. The server may set the following status codes on the data nodes:

- GOOD: normal state of operation
- BAD: measurement was not successful
- UNCERTAIN: can mean any of the following:
  - in CurrentMean and CurrentParticles: the image has too few signals resolved, data is noisy or the particle density is too high for reliable measurement
    - Check spray system status, sensor alignment and measurement parameters to ensure that good spray images are generated
  - in PlumeStatistics: the particle count is too low for good statistic. If the status does not improve over time:
    - first check the sensor alignment and other possible issues in the measurement
    - consider increasing the session length to collect more particles in the dataset.
- BAD_WaitingForInitialData: the measurement has not yet started

Particle data displaying other than GOOD status should not be relied on in determining the spray properties.

4.3 OPC UA server node structure reference

4.3.1 ROOT FOLDER

4.3.1.1 NODE ID: ns=2;s=SoftwareBuild

- **Browse name:** SoftwareBuild
- **Data type:** String
- **Access level:** Read
- **Variable description:** The minor release (build) version of the server software.

NODE ID: ns=2;s=SoftwareClass

- **Browse name:** SoftwareClass
- **Data type:** String
- **Access level:** Read
- **Variable description:** OCP UA server software identifier. This value is related to sensor models supported by the software.

NODE ID: ns=2;s=MaxSensorCount

- **Browse name:** MaxSensorCount
- **Data type:** Int32
- **Access level:** Read
- **Variable description:** Maximum nuber of sensors supported by the server.

NODE ID: ns=2;s=SoftwareVersion

- **Browse name:** SoftwareVersion
- **Data type:** String
- **Access level:** Read
- **Variable description:** The major release version of the server software.

4.3.2 FOLDER ID: ns=2;s=Sensor_00

- **Browse name:** Sensor_00
- **Folder description:** Sensor node folder. Folder internal structure and subfolder names defined by the sensor type.

NODE ID: ns=2;s=Sensor_00.Location

- **Browse name:** Location
- **Data type:** String
- **Access level:** Read
- **Variable description:** User defined location ID for the sensor device.

NODE ID: ns=2;s=Sensor_00.SensorType

- **Browse name:** SensorType
- **Data type:** String
- **Access level:** Read
- **Variable description:** Name of the sensor device model.

NODE ID: ns=2;s=Sensor_00.SensorSN

- **Browse name:** SensorSN
- **Data type:** String
- **Access level:** Read
- **Variable description:** Serial number of the sensor device.

4.3.3 FOLDER ID: ns=2;s=Sensor_00.CurrentMean

- **Browse name:** CurrentMean
- **Folder description:** Nodes in this folder have historical access enabled with history length: 1000 points.

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleSpeed

- **Browse name:** ParticleSpeed
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleSpeed

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleDensity

- **Browse name:** ParticleDensity
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleDensity

NODE ID: ns=2;s=Sensor_00.CurrentMean.SprayPos

- **Browse name:** SprayPos
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: SprayPos

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleCount

- **Browse name:** ParticleCount
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleCount

NODE ID: ns=2;s=Sensor_00.CurrentMean.SprayWidth

- **Browse name:** SprayWidth
- **Data type:** Double

- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: SprayWidth

NODE ID: ns=2;s=Sensor_00.CurrentMean.CurrentAlertMessage

- **Browse name:** CurrentAlertMessage
- **Data type:** String
- **Access level:** Read
- **Variable description:** Message specifying the most recent alert condition.

4.3.4 FOLDER ID: ns=2;s=Sensor_00.CurrentParticles

- **Browse name:** CurrentParticles
- **Folder description:** Nodes in this folder have historical access enabled with history length: 1000 points.

NODE ID: ns=2;s=Sensor_00.CurrentParticles.ParticleAxVel

- **Browse name:** ParticleAxVel
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Array format data for single particle properties from most recent image bundle: ParticleAxVel

NODE ID: ns=2;s=Sensor_00.CurrentParticles.ParticleLatVel

- **Browse name:** ParticleLatVel
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Array format data for single particle properties from most recent image bundle: ParticleLatVel

NODE ID: ns=2;s=Sensor_00.CurrentParticles.ParticleAxPos

- **Browse name:** ParticleAxPos
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Array format data for single particle properties from most recent image bundle: ParticleAxPos

NODE ID: ns=2;s=Sensor_00.CurrentParticles.ParticleLatPos

- **Browse name:** ParticleLatPos
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Array format data for single particle properties from most recent image bundle: ParticleLatPos

NODE ID: ns=2;s=Sensor_00.CurrentParticles.ParticleDiameter

- **Browse name:** ParticleDiameter
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Array format data for single particle properties from most recent image bundle: ParticleDiameter

---

4.3.5 FOLDER ID: ns=2;s=Sensor_00.PlumeStatistics

- **Browse name:** PlumeStatistics
- **Folder description:** Nodes in this folder have historical access enabled with history length: 1000 points.

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.PlumeCount

- **Browse name:** PlumeCount
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: PlumeCount

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.PeakDensityPos

- **Browse name:** PeakDensityPos
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: PeakDensityPos

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.PeakAxVelPos

- **Browse name:** PeakAxVelPos
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: PeakAxVelPos

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.PeakAxVel

- **Browse name:** PeakAxVel
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: PeakAxVel

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.SizeCount

- **Browse name:** SizeCount
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: SizeCount

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.SizeDV50

- **Browse name:** SizeDV50
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: SizeDV50

NODE ID: ns=2;s=Sensor_00.PlumeStatistics.SizeDV90

- **Browse name:** SizeDV90
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Particle collection statistic for spray plume: SizeDV90

4.3.6 FOLDER ID: ns=2;s=Sensor_00.SensorStatus

- **Browse name:** SensorStatus
- **Folder description:** Sensor status information.

NODE ID: ns=2;s=Sensor_00.SensorStatus.CalibrationDate

- **Browse name:** CalibrationDate
- **Data type:** String
- **Access level:** Read
- **Variable description:** Date of the presently loaded device calibration file.

NODE ID: ns=2;s=Sensor_00.SensorStatus.HardwareProfile

- **Browse name:** HardwareProfile

- **Data type:** String
- **Access level:** Read
- **Variable description:** Sensor hardware profile ID selected at startup.

NODE ID: ns=2;s=Sensor_00.SensorStatus.CameraState

- **Browse name:** CameraState
- **Data type:** String
- **Access level:** Read
- **Variable description:** Current operation state of the camera device.

NODE ID: ns=2;s=Sensor_00.SensorStatus.CameraTemp

- **Browse name:** CameraTemp
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Camera device internal temperature.

NODE ID: ns=2;s=Sensor_00.SensorStatus.LaserPulsing

- **Browse name:** LaserPulsing
- **Data type:** Array of Double
- **Access level:** Read
- **Variable description:** Laser pulse structure in microseconds. Empty if laser pulsing is disabled. Array items: 0: pulse count 1: pulse duration 2: pulse interval

NODE ID: ns=2;s=Sensor_00.SensorStatus.CameraGain

- **Browse name:** CameraGain
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Sensor camera device gain setting in dB

NODE ID: ns=2;s=Sensor_00.SensorStatus.FrameRate

- **Browse name:** FrameRate
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Recent sensor frame rate as frames per second.

NODE ID: ns=2;s=Sensor_00.SensorStatus.SensorErrorCode

- **Browse name:** SensorErrorCode
- **Data type:** Int32
- **Access level:** Read
- **Variable description:** Most recent error code given by the sensor

NODE ID: ns=2;s=Sensor_00.SensorStatus.SensorErrorMessage

- **Browse name:** SensorErrorMessage
- **Data type:** String
- **Access level:** Read
- **Variable description:** Error message associated with the SensorErrorCode

---

4.3.7 FOLDER ID: ns=2;s=Sensor_00.SessionData

- **Browse name:** SessionData
- **Folder description:** Measurement session related information.

NODE ID: ns=2;s=Sensor_00.SessionData.SessionName

- **Browse name:** SessionName
- **Data type:** String

- **Access level:** Read
- **Variable description:** User provided name for the measurement session

NODE ID: ns=2;s=Sensor_00.SessionData.AlertProfileList

- **Browse name:** AlertProfileList
- **Data type:** Array of String
- **Access level:** Read
- **Variable description:** Names of available sensor alert setting profiles.

NODE ID: ns=2;s=Sensor_00.SessionData.ActiveAlertProfile

- **Browse name:** ActiveAlertProfile
- **Data type:** String
- **Access level:** Read
- **Variable description:** Name of the currently active sensor alert profile.

NODE ID: ns=2;s=Sensor_00.SessionData.MaxLength

- **Browse name:** MaxLength
- **Data type:** Int32
- **Access level:** Read
- **Variable description:** Maximum count of image bundles included in the session data set.

NODE ID: ns=2;s=Sensor_00.SessionData.CurrentLength

- **Browse name:** CurrentLength
- **Data type:** Int32
- **Access level:** Read
- **Variable description:** Current count of image bundles included in the session data set.

NODE ID: ns=2;s=Sensor_00.SessionData.OutputFolder

- **Browse name:** OutputFolder
- **Data type:** String
- **Access level:** Read
- **Variable description:** Folder base path for the output files generated when session data is stored.

NODE ID: ns=2;s=Sensor_00.SessionData.LastOutputPath

- **Browse name:** LastOutputPath
- **Data type:** String
- **Access level:** Read
- **Variable description:** Absolute pathname of the folder where the last session data was stored.

4.3.8 FOLDER ID: ns=2;s=Sensor_00.OPERATIONS

- **Browse name:** OPERATIONS
- **Folder description:** Node structure for issuing commands to the sensor control software.

NODE ID: ns=2;s=Sensor_00.OPERATIONS.ENABLED

- **Browse name:** ENABLED
- **Data type:** Boolean
- **Access level:** Read
- **Variable description:** Node value informs if remote control is enabled for the sensor

NODE ID: ns=2;s=Sensor_00.OPERATIONS.COMMAND

- **Browse name:** COMMAND
- **Data type:** String
- **Access level:** Read&Write
- **Variable description:** Writable node for issuing commands to the sensor software.

NODE ID: ns=2;s=Sensor_00.OPERATIONS.RESULT

- **Browse name:** RESULT
- **Data type:** String
- **Access level:** Read
- **Variable description:** Node contains the issued command and the result code of the command processor.

NODE ID: ns=2;s=Sensor_00.OPERATIONS.USERPROMPT

- **Browse name:** USERPROMPT
- **Data type:** String
- **Access level:** Read
- **Variable description:** Node contains a display string for the user's attention.

4.3.9 FOLDER ID: ns=2;s=Sensor_00.CurrentMean

- **Browse name:** CurrentMean
- **Folder description:** Nodes in this folder have historical access enabled with history length: 1000 points.

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleSpeed

- **Browse name:** ParticleSpeed
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleSpeed

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleDensity

- **Browse name:** ParticleDensity
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleDensity

NODE ID: ns=2;s=Sensor_00.CurrentMean.SprayPos

- **Browse name:** SprayPos
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: SprayPos

NODE ID: ns=2;s=Sensor_00.CurrentMean.ParticleCount

- **Browse name:** ParticleCount
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: ParticleCount

NODE ID: ns=2;s=Sensor_00.CurrentMean.SprayWidth

- **Browse name:** SprayWidth
- **Data type:** Double
- **Access level:** Read
- **Variable description:** Most recent mean value for measurement: SprayWidth

NODE ID: ns=2;s=Sensor_00.CurrentMean.CurrentAlertMessage

- **Browse name:** CurrentAlertMessage
- **Data type:** String
- **Access level:** Read
- **Variable description:** Message specifying the most recent alert condition.

4.4 Notes for specific data nodes:

- **PlumeStatistics.SizeCount**; all sizing measurements remain inactive unless Sizing is enabled in the HMI.
- **PlumeStatistics.SizeDV50** and **PlumeStatistics.SizeDV90** require that corresponding sizing function is enabled in the HMI.
- **PlumeStatistics.PeakAxVelPos** and **PlumeStatistics.PeakAxVel** need that "Crossplots" display in the HMI has the "Lateral V" output setting enabled.
- **CurrentMean.CurrentAlertMessage** is offered for the convenience of the clients that do not support OPC UA Events. Monitoring the Events is recommended implementation of system feedback. See Section 4.5 for more information.

4.5 Alert objects

The sensor offers Alert node for monitoring user set level limits for user specified CurrentMean values. The values and limits for alert generation are defined in the HiWatch software user interface.

- **Browse name:** CurrentMeanAlertmon
- **Event description:** Nonexclusive Level Alert for all defined CurrentMean variables

The OPC UA client software can subscribe to Alert node to receive OPC UA Events when the given variable exceeds the defined range limits.

The alert settings are determined and enabled in the HiWatch HMI screen. Refer to the software documentation for details.

NOTE: The OPC UA TCP session is disconnected when alert levels are changed. The client should try to reconnect in shorrt notice. A refresh request may be necessary from the client side to avoid missing condition events during reconnect.

5 Command interface

The OPC UA server publishes a COMMAND variable node for remote control of the sensor. The client must write a String value to this variable to use the remote functionality. The RESULT node shows the processed command with result code after the processing is finished.

The ENABLED node indicates that the server will accept remote commands. Check the value of this node before attempting remote operations. The ENABLED condition is controlled by the operator of the server software.

NOTE: the result code is only related to the command syntax and structure information, not the actual successful or unsuccessful completion of the operation or its validity. The user should read the related SensorStatus node to verify the outcome of the command execution.

5.1 Command structure

The ASCII command string value written to the OPC UA node COMMAND must have the following structure:

cmdstring {*parameter*}

cmdstring is one of the implemented commands, optional *parameter* is determined by the command specification.

5.2 Available commands

The SprayWatch OPC UA server implements the following user commands:

- SENSOR: controls the operation of the sensor
  - deprecated CAMERA command is still available and works with the same parameters

- PULSE: controls the pulsing state of the strobe laser
- SESSION: controls the remote data handling and storage operations

The usage, relevant parameters and options for each command are listed below.

5.2.1 SENSOR

Syntax:

SENSOR [ACTIVE|IDLE]

Sets camera state to Active or Idle as specified by the mandatory parameter.

NOTE: Automatic state changes or manual mode switch by the user are still possible.

The current state of the camera is observed with **SensorStatus.CameraState** node. Before issuing SENSOR ACTIVE command, check that the state is Standby. In case of any other status code, the state of the sensor must be corrected before SENSOR ACTIVE will be successful.

SENSOR IDLE command will work only when the status is Active, the sensor will enter Standby state when completed.
NOTE: Sensor will exit Active state if operating conditions

5.2.2 PULSE

Syntax:

PULSE [CT|DUR|INT] [*value (floating point)*]

Adjusts the laser strobe pulsing parameter listed. Other parameters will remain untouched. One *value* parameter must be given. The value uses point (.) as the decimal separator.

Switches (exactly one must be selected):

- CT: Sets the number of strobe pulses per frame to *value* (rounded down).
- DUR: Sets the time duration of strobe pulses *value* in microseconds.
- INT: Sets the time interval of strobe pulses *value* in microseconds.

Any value will be accepted in the command. Configuring the strobe laser will be attempted with the given value, however the actual result may be different. Read the **LaserPulsing** node for information of the current pulsing values. Up to 1 s delay may be expected before the **PULSE** command takes effect and is shown in the **LaserPulsing** node.

NOTE: HiWatch HR2 sensor will accept only INT switch for this command.

5.2.3 SESSION

Syntax: SESSION [RESET|STORE|FOLDER|MAXLENGTH] [*parameter*]

Switches explained:

- SESSION RESET: clears the current particle dataset at the start of new measurement session.
- SESSION STORE [*name*]: Stores the current particle dataset to the Data table in the HMI, and to a file in the Autosave folder. The parameter *name* must be
  - printable US-ASCII string (no control characters)
  - maximum 64 bytes length
  - must contain no whitespace.
    The name of the output file will also include the Location parameter set for the sensor.
    NOTE: the user must monitor the available storage space for new datasets.
- SESSION FOLDER [*foldername*]: defines the output folder for the session autosave data. The parameter *foldername* must be
  - printable US-ASCII string (no control characters)
  - maximum 256 bytes length

    - must be valid Windows pathname, to a path writable by the HMI software
    - must contain no whitespace, unless entirely in double quotes (")
- SESSION MAXLENGTH [number]: sets the maximum count of image bundles included in the particle data collection.
  The data count will be zero after SESSION RESET command and increase when new particle data is available. If the MAXLENGTH value is exceeded, the oldest data in the particle dataset will be dropped.

The success of any SESSION command should be verified by checking the value of any corresponding SessionData node after issuing the command. The nodes for respective switches include:

- RESET: **SessionStatus.CurrentLength** should go to zero until more particle data arrives
- STORE: **SessionStatus.SessionName**, **SessionStatus.LastOutputPath** will be properly set
- FOLDER: **SessionStatus.OutputFolder** will be properly set
- MAXLENGTH: **SessionStatus.MaxLength** will be properly set

In addition, **SessionStatus.ActiveAlertProfile** will show if there is an active level alert profile that has been enabled from the HMI.

Limitations

The OPC UA server offers no support for the following features of OPC UA standard:

- methods
- views

Any equivalent functionality must be implemented by the user in the client software.

6 Installing HiWatch OPC UA HMI software

Installation of HiWatch OPC UA software is very similar to the standard version, with the following additional steps:

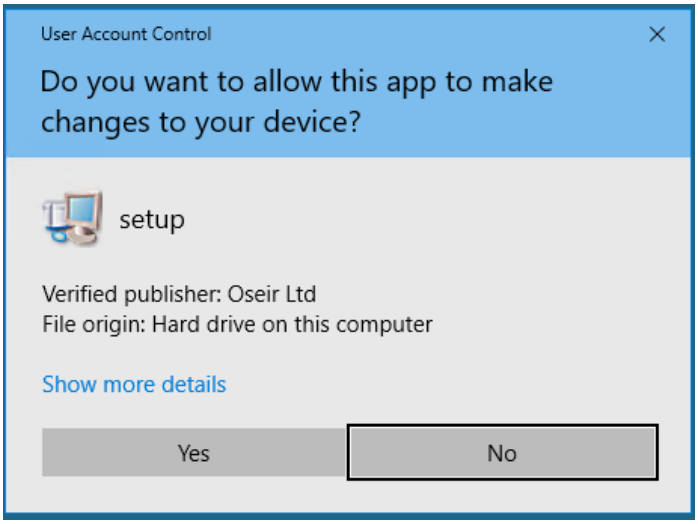
- activation of OPC UA server license
- granting permission of OPC UA network traffic in the system firewall.

6.1 Running the HiWatch OPC UA installer

The HiWatch OPC UA software is delivered on a physical medium with the HiWatch system, or a downloadable upgrade in compressed format. The compressed ZIP archive must be completely extracted to a local disk before starting the upgrade. **The upgrade will work only on a system that has a working installation of HiWatch software 3.5 or later.**

The installation/upgrade is started by running **setup.exe** program on the delivery media. The installing user must have administration privileges on the target system.

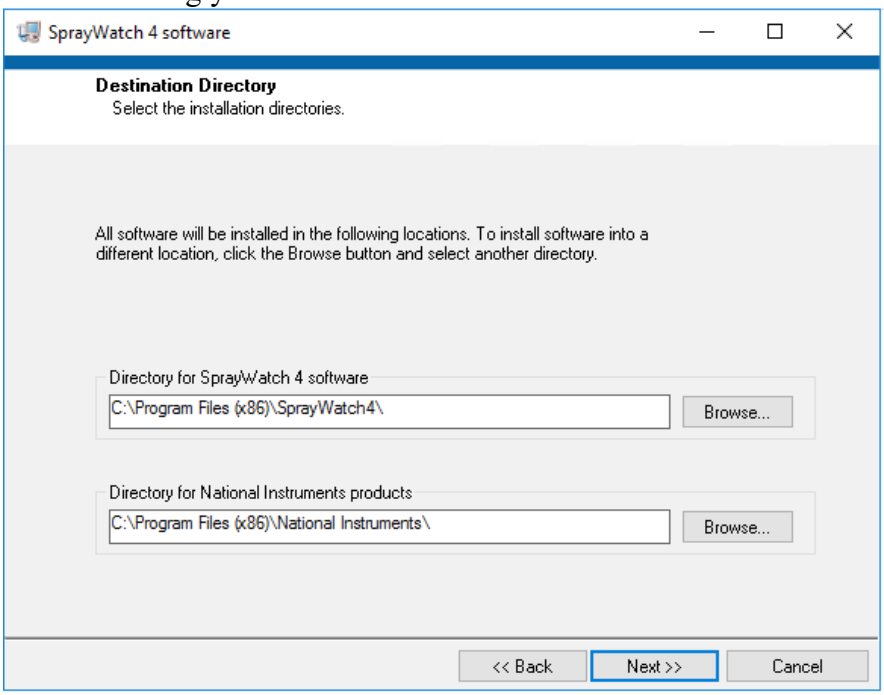
Starting from version 1.1, all HiWatch software will be digitally signed to enhance software security. Please verify that the User Account Control popup has ”Oseir Ltd” as the ”Verified publisher”.

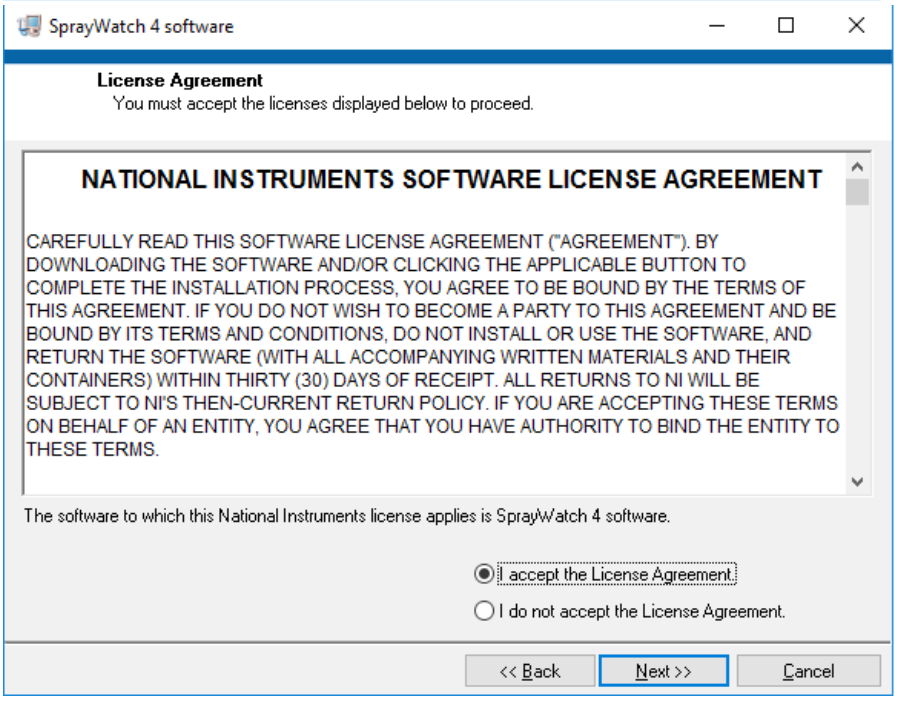


After accepting the privileges of the installer, the following choices must be made:

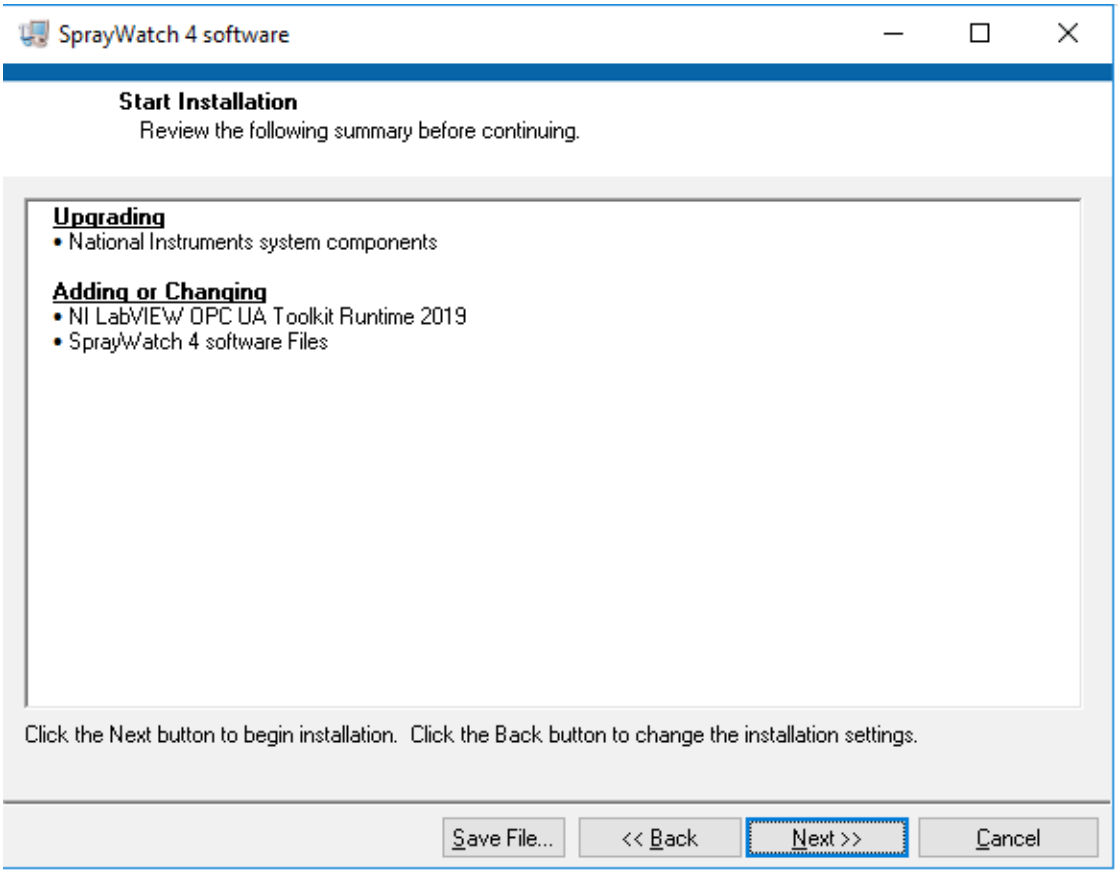
- software destination
- accepting the related NI licenses

The default values are strongly recommended for destination choices.

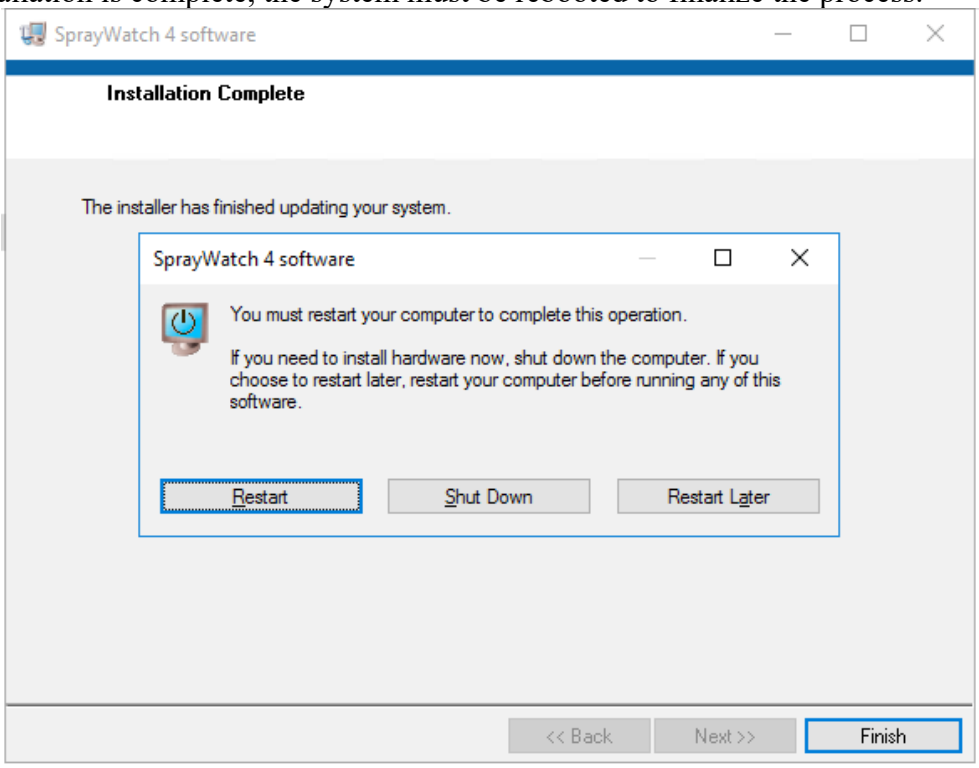




The upgrade installer will add new version of HiWatch software and OPC UA toolkit as below:



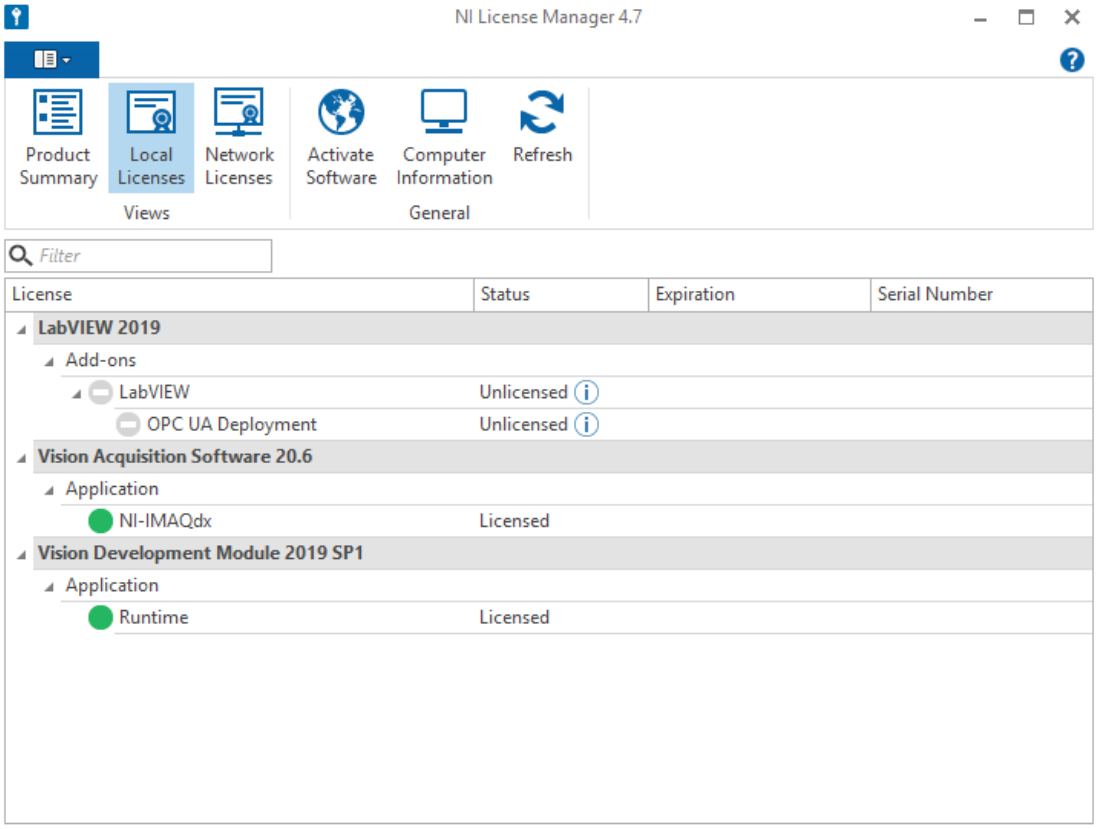
After the installation is complete, the system must be rebooted to finalize the process.



6.2 Activation of OPC UA server license

The final step of the upgrade is to activate the OPC UA server software.

Run ”NI License Manager” from the system menu, and select ”Local Licenses”:



The actual components displayed may vary according to the local software installed.

The ”OPC UA Deployment” item must be activated. For that, the user will need

- unique ”Computer ID” of the target system
- serial number of the license, supplied with the system or upgrade delivery.

The Computer ID is revealed by clicking the ”Computer Information” item.

The activation can be made at the Web address:

https://www.ni.com/nilg-activate/jsp/customer_activate_details.jsp

This can be done in another computer. The link contains a Web form that will ask for:

- Computer OS version
- Activated software component name and version
- License Serial number
- Destination Computer ID
- some personal information, but its veracity may not be checked.

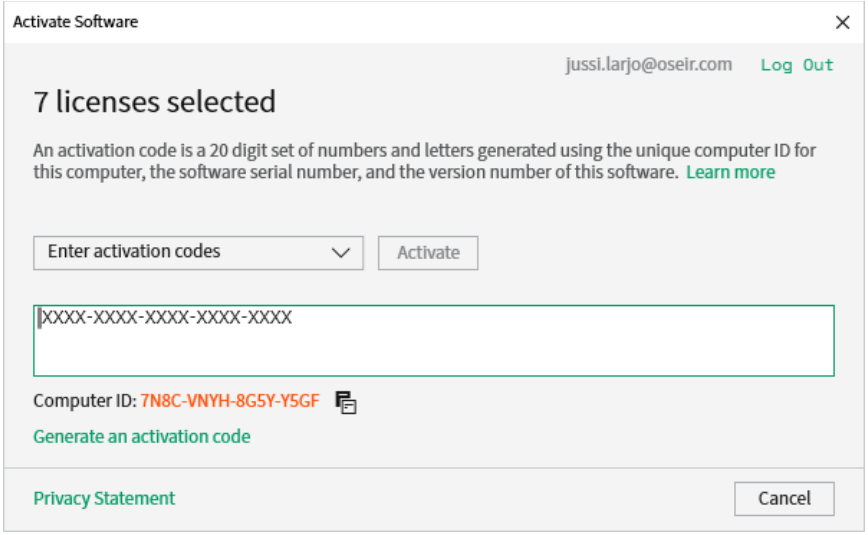
Entering a valid email address may be convenient, as a copy of the activation information will be sent to that address.

For HiWatch software version 3.5, use the following information:

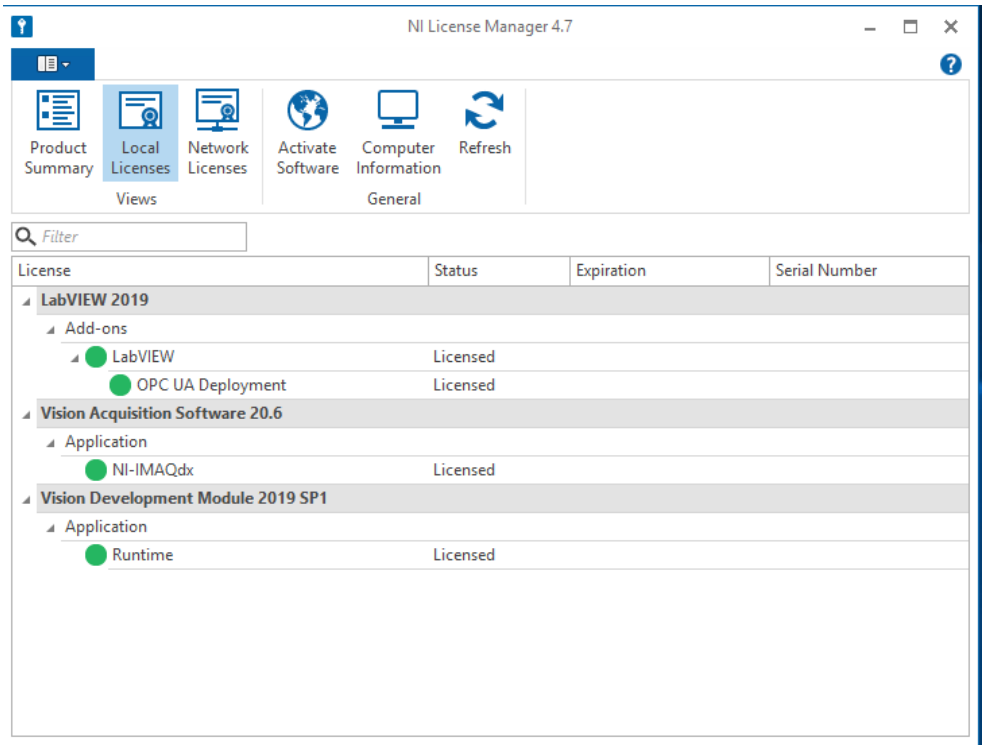
`Product:` `OPC UA Deployment`

After entering the necessary details correctly, the following Web page will reveal the correct Activation Code for the software.

Next, click ”Activate software” on the License manager. In the popup window, select ”Enter activation codes” method.



Enter the 20-character activation code to the box and click ”Activate” button. The activated component will be selected automatically.



Finally, the user may check that the ”OPC UA Deployment” component is correctly activated. If positive, the SprayWatch OPC UA Server is now functional.

6.3 Running HiWatch OPC UA server

After correct upgrade and activation, the system menu will contain following items:

After the upgrade, the original HiWatch HMI software version will still be available and can be used without OPC UA activation. Select the CS2/HR2_OPCUA item to start the OPC UA version.

When running the OPC UA version for the first time, the system may require opening of the firewall. In a typical case, the service should be allowed for both Private and Public networks for correct operation.

The final step is to connect a HiWatch sensor to the software and load the initial profile for it. After that the user may check that the correct software version is running and the OPC server URL is displayed on the Remote page (Fig.1).

After this, the user may connect any compatible OPC UA client software using the displayed server URL and obtain the required data using the connection during a measurement session.

© Oseir Ltd. 2026